



AMC ENGINEERING COLLEGE

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

Capacity Planning for IT

Sub Code:BCS456B

MODULE-2

1. which of the following used in distributed monitoring software?

- A) Gomez
- B) Ganglia
- C) Elastra
- D) None of the above.

Answer-B

2. What is the primary purpose of capacity tracking tools?

- A) To generate alerts concerning system availability
- B) To automatically and routinely measure server behavior over a predefined amount of time
- C) To manage network configurations
- D) To install and update software on servers

Answer-B To automatically and routinely measure server behavior over a predefined amount of time

3. Which of the following tools is NOT typically used for load testing?

- A) Loadrunner
- B) JMeter
- C) Iago
- D) Grafana

Answer- D) Grafana

4.What is a common database format used by metric aggregation servers to store time-series data?

- A) SQL Database
- B) NoSQL Database
- C) Round-Robin Database (RRD)
- D) Hierarchical Database

Answer- C) Round-Robin Database (RRD)

5.Which tool is specifically designed for querying and visualizing time-series operational metrics?

- A) Ganglia
- B) Graphite
- C) Grafana
- D) Cacti

Answer C) Grafana

6.Which of the following is a common daemon used to collect system and application performance metrics periodically?

- A) Riemann
- B) Datadog
- C) collectd
- D) Sematext

Answer C) collectd

7.What is the main advantage of using RRDTool for storing system and network data?

- A) It can store unlimited data without any loss

- B) It assumes you are interested in fine details only for the recent past
- C) It is a real-time data analysis tool
- D) It is primarily used for alert generation

Answer B) It assumes you are interested in fine details only for the recent past

8.What is the role of the agent in most metric collection systems?

- A) To display the metrics in various forms
- B) To periodically collect data from the machine on which it's running and send a summary to the metric aggregation server
- C) To manage network configurations
- D) To handle user authentication and authorization

Answer **B) To periodically collect data from the machine on which it's running and send a**

9.Why might storing long periods of operations data be not useful, according to the chapter?

- A) It can be costly
- B) It can misguide decision-making due to data staleness
- C) It is illegal to store data for long periods
- D) It takes too much time to analyze

Answer B) It can misguide decision-making due to data staleness

10.What protocol is commonly used for gathering metrics for most networking and server equipment?

- A) HTTP
- B) FTP
- C) SNMP
- D) TCP/IP

Answer) SNMP

11.What is the primary reason for using load balancers in web operations?

- A) To establish secure connections
- B) To distribute load among pools or clusters of machines
- C) To manage database queries
- D) To monitor network traffic

Answer B) To distribute load among pools or clusters of machines

12.What is the primary reason for collecting application-level metrics?

- A) To monitor CPU and disk usage on a single server
- B) To provide a comprehensive view of the entire system's performance
- C) To measure network latency
- D) To track the number of concurrent users on a website

Answer: B) To provide a comprehensive view of the entire system's performance

13.Which of the following is NOT an example of an application-level metric mentioned in the text?

- A) Number of Tweets per minute
- B) Number of Messages per minute
- C) CPU usage per server
- D) Number of Photos uploaded per minute

Answer: C) CPU usage per server

14.What is one of the main benefits of tracking application-level metrics over server-specific metrics?

- A) They are easier to collect
- B) They provide insight into business and financial impacts
- C) They are less prone to errors
- D) They require less storage space

Answer: B) They provide insight into business and financial impacts

15. Which storage analogy is used in the text to explain data storage considerations?

- A) A bucket of sand
- B) A box of chocolates
- C) A glass of water
- D) A balloon

Answer: C) A glass of water

16. What storage type is described as having "large, slow, inexpensive disks"?

- A) SSDs
- B) SAS drives
- C) SCSI drives
- D) ATA/SATA drives

Answer: D) ATA/SATA drives

17. Which metric is NOT typically used to compare the performance of different HDDs?

- A) Sustained transfer rate
- B) Average latency
- C) Network bandwidth
- D) Cache buffer size

Answer: C) Network bandwidth

18. What key metric should be tracked to predict when more storage hardware needs to be ordered?

- A) Total disk space consumption over time
- B) Number of file accesses per minute
- C) Disk I/O operations per second
- D) CPU utilization on storage servers

Answer: A) Total disk space consumption over time

19. In the real-world example provided, which days were identified as peak periods for photo uploads at Flickr?

- A) Fridays
- B) Mondays
- C) Sundays
- D) Saturdays

Answer: C) Sundays

20. What is the main reason for setting up retention policies for long-term data storage?

- A) To increase data retrieval speed
- B) To reduce storage costs
- C) To improve data accuracy
- D) To enhance server performance

Answer: B) To reduce storage costs

21. Which tool is NOT mentioned as being used for measuring storage performance?

- A) Hard Disk Sentinel
- B) IOBench
- C) ATTO Disk Benchmark
- D) Wireshark

Answer: D) Wireshark

22. Based on the provided text, which of the following are important metrics to track with any caching software?

- A) Total server uptime
- B) Database query execution time
- C) Cache hit ratio
- D) Number of CPU cores

Correct answer: C) Cache hit ratio

23. What defines the capacity ceiling for a cache with a small or slowly increasing working set?

- a) CPU usage
- b) Hit ratio
- c) Disk I/O utilization
- d) Network usage

Correct Answer: b) Hit ratio

24. In the context of caching systems, what is used as a guide to determine the maximum performance of the hardware platform?

- a) Memory usage
- b) Cache size
- c) Hit ratio
- d) Disk I/O wait

Correct Answer: c) Hit ratio

25. What system resource metric is identified as the defining peak resource metric for a Squid caching server?

- a) CPU usage
- b) Memory usage
- c) Disk I/O wait

d) Network usage

Correct Answer: c) Disk I/O wait

26. In the scenario discussed, what is the relationship between photo processing rates and CPU utilization?

- a) Linear
- b) Exponential
- c) Inverse
- d) No correlation

Correct Answer: a) Linear

27. What is the purpose of rate-limiting API calls in web services?

- a) To increase API usage
- b) To decrease API usage
- c) To prevent abuse and ensure fair usage
- d) To track API usage for billing purposes

Correct Answer: c) To prevent abuse and ensure fair usage

VTUSYNC.IN